



# Technical Service Bulletin

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New Turbocharger Oil Supply Line Male Union, QSK19 Engine

## New Turbocharger Oil Supply Line Male Union, QSK19 Engine

### Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

### Contents

This document introduces new turbocharger lubricating oil plumbing hardware for the QSK19 engine. A new male union, Part Number 2888894, was released to address turbocharger plumbing routing issues. This document provides resolution to the issue described in TSB110039. Table 1 shows the new part number and o-rings included in this assembly.

**Table 1: New Male Union and O-rings**

| Part Number | Description  | Available |
|-------------|--------------|-----------|
| 2888894     | Union, male  | Yes       |
| Contains:   |              |           |
| 2888849     | Union, male  | No        |
| 3037537     | Seal, o-ring | Yes       |
| 3922794     | Seal, o-ring | Yes       |

The new male union is taller than the previous unions used at the turbocharger oil inlet. This allows for more clearance between the turbocharger oil supply hose and other objects in this area, including the bearing housing and coolant vent line. Figure 1 shows the old fitting on the left and the new fitting on the right.

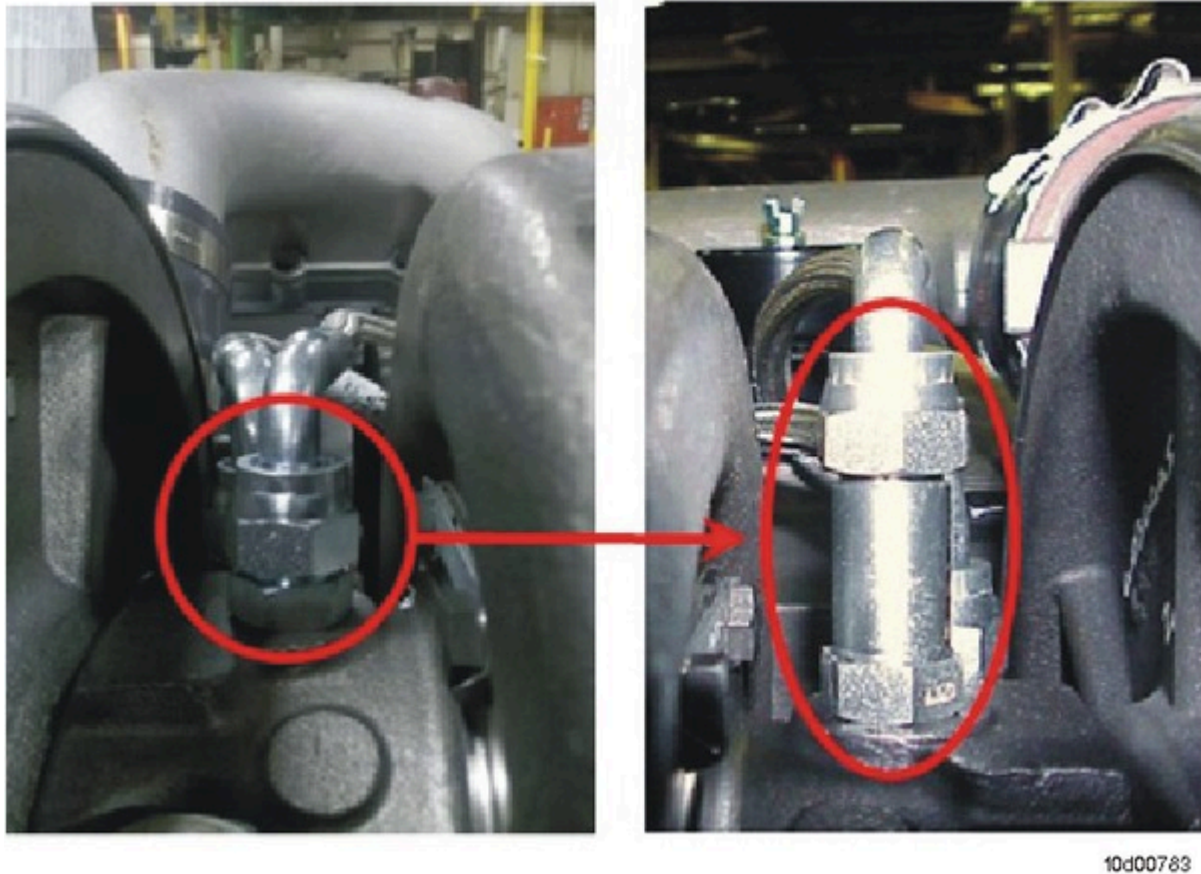


Figure 1. Old Male Union Versus New Taller Male Union

The engine serial number (ESN) first for this change is 37249951, built on 25-June-2011.

Procedures 20-010-041-tr and 20-010-046 in the QSK19 Service Manual, Bulletin 4021595, have been updated to include notification of the taller union and checks for fretting and other damage on the braided lines.

Procedures 13-019-390 in the QSK19 (CM850 Modular Common Rail System) Electronic Control System Troubleshooting and Repair Manual, Bulletin 4021493, and procedure 123-019-390 in the QSK19 CM2150 Electronic Control System Troubleshooting and Repair Manual, Bulletin 4022094, have been updated to include removal of the bracket and installation of the clips to attach the speed sensor wire to the oil supply line.

The new male union, Part Number 2888894, does **not** supersede any part numbers because these unions are used on other locations and engines. Table 2 shows the options affected by this change.

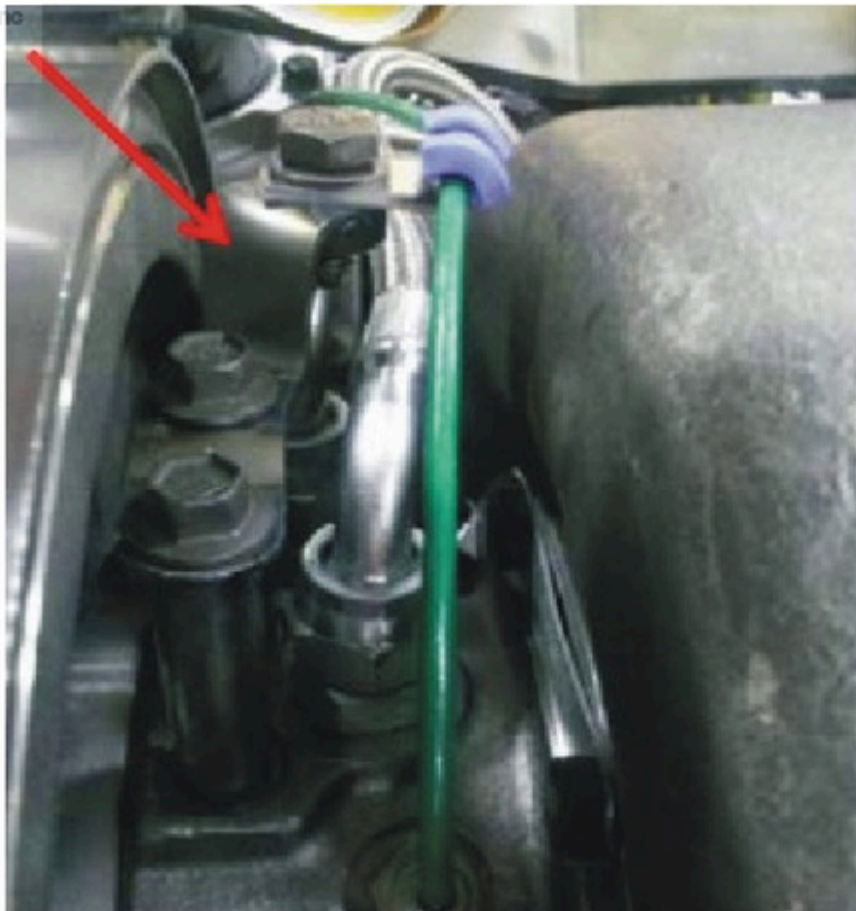
**Table 2. Options Containing a New Fitting**

| Option Number | Description            |
|---------------|------------------------|
| TP4703        | Plumbing, turbocharger |
| TP4704        | Plumbing, turbocharger |
| TP4707        | Plumbing, turbocharger |
| TP4708        | Plumbing, turbocharger |
| TP4709        | Plumbing, turbocharger |

| Table 2. Options Containing a New Fitting |                        |
|-------------------------------------------|------------------------|
| Option Number                             | Description            |
| TP4710                                    | Plumbing, turbocharger |
| TP4718                                    | Plumbing, turbocharger |
| TP4725                                    | Plumbing, turbocharger |
| TP4726                                    | Plumbing, turbocharger |
| TP4728                                    | Plumbing, turbocharger |
| TP4729                                    | Plumbing, turbocharger |
|                                           |                        |

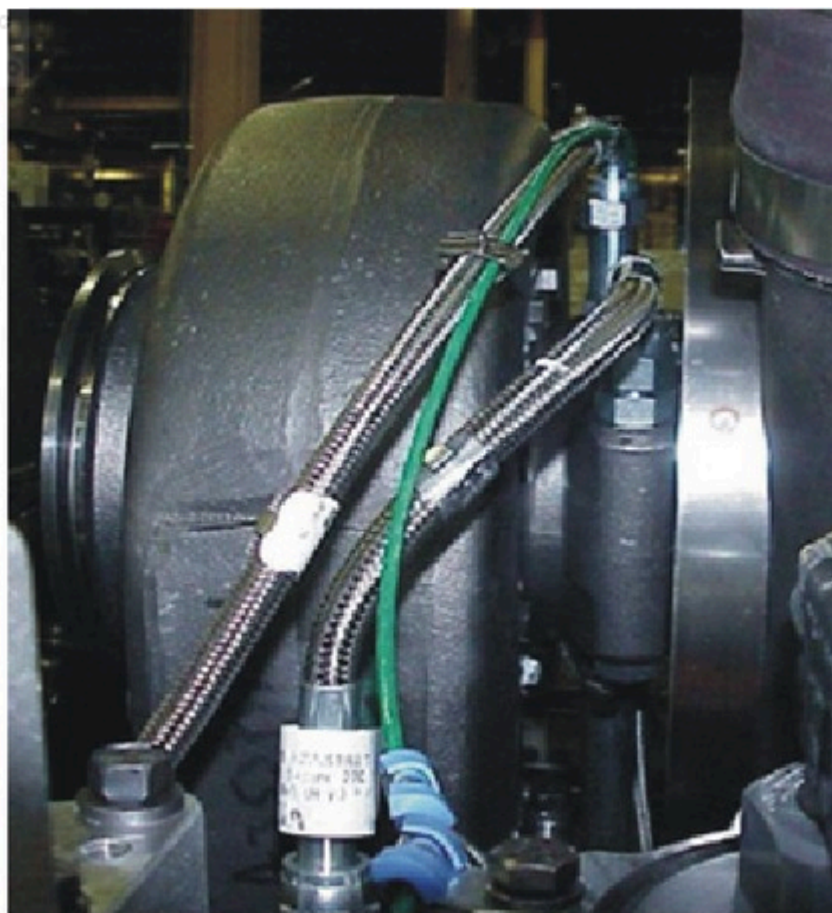
On QSK19 MCRS Tier 3 options with a turbocharger speed sensor, this new male union causes an interference with the bracket supporting the speed sensor wire. To resolve this, the bracket has been obsoleted and the hardware has been removed from the affected options. Now, the wire should be clipped to the oil supply line to allow for maximum clearance. This is similar routing to other turbocharger options. Table 3 shows the structure changes to these options. Figure 2 shows the bracket that is being obsoleted. Figure 3 shows the new turbocharger speed sensor wire routing. Table 4 shows the options that had the bracket removed.

| Table 3: Added and Removed Part Numbers from Affected Options |     |                 |                 |     |                            |
|---------------------------------------------------------------|-----|-----------------|-----------------|-----|----------------------------|
| Added                                                         |     |                 | Removed         |     |                            |
| New Part Number                                               | Qty | Description     | Old Part Number | Qty | Description                |
| 4924789                                                       | 2   | Clamp, Wire Tie |                 |     |                            |
|                                                               |     |                 | 4924749         | 1   | Bracket, Wiring Retainer   |
|                                                               |     |                 | 3093927         | 2   | Screw, Hex Flange Head Cap |
|                                                               |     |                 | 4924291         | 1   | Clip                       |
|                                                               |     |                 | 3015283         | 1   | Screw, Captive Washer Cap  |
|                                                               |     |                 |                 |     |                            |



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Figure 2. Previous Speed Sensor Wire Routing with Bracket



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Figure 3. New Speed Sensor Wire Routing with Clips

| Table 4. Options With Bracket Removed |                                     |
|---------------------------------------|-------------------------------------|
| Option                                | Description                         |
| PH4760                                | Mounting, turbocharger speed sensor |
| PH4761                                | Mounting, turbocharger speed sensor |
| PH4788                                | Mounting, turbocharger speed sensor |
| PH4789                                | Mounting, turbocharger speed sensor |

## Document History

| Date      | Details        |
|-----------|----------------|
| 2011-6-23 | Module Created |